

Supplemental Materials: Successive Height Preintegration for a Height-Interdependent Path Formulation of Multiple Bounces in Smith Microfacet BRDFs

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In this document, we provide additional details on the segment term in our model, as well as further details and validation of PTABE.

CCS Concepts: • **Computing methodologies** → **Rendering; Reflectance modeling.**

ACM Reference Format:

Siyuan Zhang, Takuya Funatomi, Yuki Fujimura, Yasuhiro Mukaigawa, and Hiroyuki Kubo. 2026. Supplemental Materials: Successive Height Preintegration for a Height-Interdependent Path Formulation of Multiple Bounces in Smith Microfacet BRDFs. In *Special Interest Group on Computer Graphics and Interactive Techniques Conference Conference Papers (SIGGRAPH Conference Papers '26)*, July 19–23, 2026, Los Angeles, CA, USA. ACM, New York, NY, USA, 20 pages. <https://doi.org/10.1145/3799902.3811194>

1 SEGMENT TERM

In this section, we provide the explicit formulation of the height-interdependent segment term S_k , present its position-free version S_{p_k} obtained after successive preintegration over height, and prove the reciprocity of our segment term.

1.1 Height-interdependent segment term

In the main paper, given the definitions of $S_k^{t_1}$ (Eq. 18 in the main paper), $S_k^{t_2}$ (Eq. 24 in the main paper), and $S_k^{t_3}$ (Eq. 25 in the main paper), we can express the height-interdependent S_k as follows:

$$S_k = S_k^{\text{local}}(\mathbf{d}_0, \dots, \mathbf{d}_k) S_k^{\text{dist}}(\mathbf{d}_0, h_0, \dots, h_{k-1}, \mathbf{d}_k), \quad (1)$$

where the backfacing microfacets are discarded in the local segment term $S_k^{\text{local}} = \prod_{i=0}^{k-1} G_1^{\text{local}}(-\mathbf{d}_i, \frac{-\mathbf{d}_i + \mathbf{d}_{i+1}}{\|-\mathbf{d}_i + \mathbf{d}_{i+1}\|})$. Since the local term is independent of height and does not participate in the subsequent derivation, we separate it from S_k for convenience. S_k^{dist} denotes the distant segment term that accounts for the effect of distant points on the microsurface.

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We provide the explicit formulation of S_k^{dist} here:

$$S_k^{\text{dist}} = \begin{cases} G_1^{\text{dist}}(-\mathbf{d}_0, h_0) \left(\prod_{i=1}^{k-1} G_1^{\text{dist}}(-\mathbf{d}_i, h_i, h_{i-1}) \right) G_1^{\text{dist}}(\mathbf{d}_k, h_{k-1}), & \text{if } \mathbf{d}_{k-1} \cdot \mathbf{d}_k < 0, \\ G_1^{\text{dist}}(\mathbf{d}_k, h_{k-1}) \left(\prod_{i=k-1}^1 G_1^{\text{dist}}(\mathbf{d}_i, h_{i-1}, h_i) \right) G_1^{\text{dist}}(-\mathbf{d}_0, h_0), & \text{if } \mathbf{d}_0 \cdot \mathbf{d}_1 < 0, \\ \underbrace{G_1^{\text{dist}}(\mathbf{d}_k, h_{k-1})}_{w_1} \underbrace{G_1^{\text{dist}}(-\mathbf{d}_0, h_0) S_{k-1}^{\text{dist}}(\mathbf{d}_1, h_1, \dots, h_{k-1}, \mathbf{d}_k)}_{\text{forward}} + \\ \underbrace{G_1^{\text{dist}}(-\mathbf{d}_0, h_0)}_{w_2} \underbrace{G_1^{\text{dist}}(\mathbf{d}_k, h_{k-1}) S_{k-1}^{\text{dist}}(-\mathbf{d}_{k-1}, h_{k-2}, \dots, h_0, -\mathbf{d}_0)}_{\text{inverse}}, & \text{otherwise,} \end{cases} \quad (2)$$

where S_k^{dist} is defined for three path types. When $\mathbf{d}_{k-1} \cdot \mathbf{d}_k < 0$, S_k^{dist} corresponds to *path type 1* and is the distant form of $S_k^{t_1}$. When $\mathbf{d}_0 \cdot \mathbf{d}_1 < 0$, it corresponds to *path type 2* and is the distant form of $S_k^{t_2}$. Otherwise, it corresponds to *path type 3* and is the distant form of $S_k^{t_3}$. The definition of S_k^{dist} yields a distant segment term containing two recursive terms, which eventually terminate in *path type 2*, thereby completing the computation.

1.2 Position-free segment term

In the main paper, we propose a successive height preintegration method to eliminate the height dimension of S_k (Eq. 1), making it position-free. The resulting position-free segment term S_{p_k} is defined as:

$$S_{p_k}(\mathbf{d}_0, \dots, \mathbf{d}_k) = S_k^{\text{local}}(\mathbf{d}_0, \dots, \mathbf{d}_k) S_{p_k}^{\text{dist}}(\mathbf{d}_0, \dots, \mathbf{d}_k), \quad (3)$$

where $S_k^{\text{local}}(\mathbf{d}_0, \dots, \mathbf{d}_k) = \prod_{i=0}^{k-1} G_1^{\text{local}}(-\mathbf{d}_i, \frac{-\mathbf{d}_i + \mathbf{d}_{i+1}}{\|\mathbf{d}_i + \mathbf{d}_{i+1}\|})$. Here, $S_{p_k}^{\text{dist}}(\mathbf{d}_0, \dots, \mathbf{d}_k)$ denotes the distant position-free segment term. It is the preintegrated form of S_k^{dist} (Eq. (2)) and can be derived as follows:

$$S_{p_k}^{\text{dist}} = \begin{cases} \underbrace{\int_{-\infty}^{\infty} \int_{-\infty}^{h_0} \dots \int_{-\infty}^{h_{k-2}} S_k^{\text{dist}} P_{h_{k-2}}^1(h_{k-1}) dh_{k-1} \dots P_{h_0}^1(h_1) dh_1 P_{+\infty}^1(h_0) dh_0}_{M_{k-1}} & \text{if } \mathbf{d}_{k-1} \cdot \mathbf{d}_k < 0, \\ \underbrace{\int_{-\infty}^{\infty} \int_{-\infty}^{h_{k-1}} \dots \int_{-\infty}^{h_1} S_k^{\text{dist}} P_{h_1}^1(h_0) dh_0 \dots P_{h_{k-1}}^1(h_{k-2}) dh_{k-2} P_{+\infty}^1(h_{k-1}) dh_{k-1}}_{M_0} & \text{if } \mathbf{d}_0 \cdot \mathbf{d}_1 < 0, \\ \underbrace{G_1^{\text{dist}}(\mathbf{d}_k)}_{w_1} \underbrace{G_1^{\text{dist}}(-\mathbf{d}_0) S_{p_{k-1}}^{\text{dist}}(\mathbf{d}_1, \dots, \mathbf{d}_k)}_{\text{forward}} + \underbrace{G_1^{\text{dist}}(-\mathbf{d}_0)}_{w_2} \underbrace{G_1^{\text{dist}}(\mathbf{d}_k) S_{p_{k-1}}^{\text{dist}}(-\mathbf{d}_{k-1}, \dots, -\mathbf{d}_0)}_{\text{inverse}}, & \text{otherwise,} \end{cases} \quad (4)$$

where $S_{p_k}^{\text{dist}}$ is defined for three path types. When $\mathbf{d}_{k-1} \cdot \mathbf{d}_k < 0$, $S_{p_k}^{\text{dist}}$ corresponds to *path type 1* and is the distant form of M_0 (Eq. 29 in the main paper). More specifically,

$$\int_{-\infty}^{h_{k-2}} S_k^{\text{dist}} P_{h_{k-2}}^1(h_{k-1}) dh_{k-1}$$

corresponds to the distant form of M_{k-1} (Eq. 29 in the main paper), while

$$\int_{-\infty}^{h_0} \cdots \int_{-\infty}^{h_{k-2}} S_k^{\text{dist}} P_{h_{k-2}}^1(h_{k-1}) dh_{k-1} \cdots P_{h_0}^1(h_1) dh_1$$

corresponds to the distant form of M_1 (Eq. 29 in the main paper). Note that only the distant form of M_0 (Eq. 29 in the main paper) is the final result of interest, as it represents height preintegration applied to all bounces in reverse order. When $\mathbf{d}_0 \cdot \mathbf{d}_1 < 0$, $S_{p_k}^{\text{dist}}$ corresponds to *path type 2* and follows the same procedure as *path type 1*. Otherwise, it corresponds to *path type 3*. For *path type 3*, the two height-dependent masking terms $G_1^{\text{dist}}(-\mathbf{d}_0, h_0)$ and $G_1^{\text{dist}}(\mathbf{d}_k, h_{k-1})$ in S_k^{dist} are height-independent of the recursive terms. Therefore, they can be preintegrated independently of the two recursive terms, resulting in $G_1^{\text{dist}}(-\mathbf{d}_0)$ and $G_1^{\text{dist}}(\mathbf{d}_k)$. $S_{p_k}^{\text{dist}}$ has a closed-form solution:

$$S_{p_k}^{\text{dist}} = \begin{cases} \prod_{i=0}^{k-1} \frac{1}{1 + \Lambda(-\mathbf{d}_i) + \Lambda(\mathbf{d}_k)}, & \text{if } \mathbf{d}_{k-1} \cdot \mathbf{d}_k < 0, \\ \prod_{i=1}^k \frac{1}{1 + \Lambda(\mathbf{d}_i) + \Lambda(-\mathbf{d}_0)}, & \text{if } \mathbf{d}_0 \cdot \mathbf{d}_1 < 0, \\ \frac{1}{(1 + \Lambda(-\mathbf{d}_0))(1 + \Lambda(\mathbf{d}_k))} \left(S_{p_{k-1}}^{\text{dist}}(\mathbf{d}_1, \dots, \mathbf{d}_k) + S_{p_{k-1}}^{\text{dist}}(-\mathbf{d}_{k-1}, \dots, -\mathbf{d}_0) \right), & \text{otherwise,} \end{cases} \quad (5)$$

where we provide a detailed derivation of $S_{p_3}^{\text{dist}}$ of the path with three bounces to illustrate how the closed-form solution of $S_{p_k}^{\text{dist}}$ can be derived:

path type 1.

$$\begin{aligned} S_{p_3}^{\text{dist}}(\mathbf{d}_0, \mathbf{d}_1, \mathbf{d}_2, \mathbf{d}_3) &= \int_{-\infty}^{+\infty} \int_{-\infty}^{h_0} \int_{-\infty}^{h_1} S_3^{\text{dist}}(\mathbf{d}_0, h_0, \mathbf{d}_1, h_1, \mathbf{d}_2, h_2, \mathbf{d}_3) P_{h_1}^1(h_2) dh_2 P_{h_0}^1(h_1) dh_1 P_{+\infty}^1(h_0) dh_0 \\ &= \int_{-\infty}^{+\infty} G_1^{\text{dist}}(-\mathbf{d}_0, h_0) \int_{-\infty}^{h_0} G_1^{\text{dist}}(-\mathbf{d}_1, h_1, h_0) \underbrace{\int_{-\infty}^{h_1} G_1^{\text{dist}}(-\mathbf{d}_2, h_2, h_1) G_1^{\text{dist}}(\mathbf{d}_3, h_2) P_{h_1}^1(h_2) dh_2 P_{h_0}^1(h_1) dh_1}_{M_2} P_{+\infty}^1(h_0) dh_0 \\ &\quad \underbrace{\hspace{10em}}_{M_1} \\ &\quad \underbrace{\hspace{10em}}_{M_0} \\ &= \int_{-\infty}^{+\infty} (C^1(h_0))^{\Lambda(-\mathbf{d}_0)} \int_{-\infty}^{h_0} \left(\frac{C^1(h_1)}{C^1(h_0)} \right)^{\Lambda(-\mathbf{d}_1)} \int_{-\infty}^{h_1} \left(\frac{C^1(h_2)}{C^1(h_1)} \right)^{\Lambda(-\mathbf{d}_2)} (C^1(h_2))^{\Lambda(\mathbf{d}_3)} P_{h_1}^1(h_2) dh_2 P_{h_0}^1(h_1) dh_1 P_{+\infty}^1(h_0) dh_0 \\ &= \left(\frac{1}{1 + \Lambda(-\mathbf{d}_2) + \Lambda(\mathbf{d}_3)} \right) \int_{-\infty}^{+\infty} (C^1(h_0))^{\Lambda(-\mathbf{d}_0)} \int_{-\infty}^{h_0} \left(\frac{C^1(h_1)}{C^1(h_0)} \right)^{\Lambda(-\mathbf{d}_1)} (C^1(h_1))^{\Lambda(\mathbf{d}_3)} P_{h_0}^1(h_1) dh_1 P_{+\infty}^1(h_0) dh_0 \\ &= \left(\frac{1}{1 + \Lambda(-\mathbf{d}_2) + \Lambda(\mathbf{d}_3)} \right) \left(\frac{1}{1 + \Lambda(-\mathbf{d}_1) + \Lambda(\mathbf{d}_3)} \right) \int_{-\infty}^{+\infty} (C^1(h_0))^{\Lambda(-\mathbf{d}_0)} (C^1(h_0))^{\Lambda(\mathbf{d}_3)} P_{+\infty}^1(h_0) dh_0 \\ &= \frac{1}{(1 + \Lambda(-\mathbf{d}_0) + \Lambda(\mathbf{d}_3))(1 + \Lambda(-\mathbf{d}_1) + \Lambda(\mathbf{d}_3))(1 + \Lambda(-\mathbf{d}_2) + \Lambda(\mathbf{d}_3))}. \end{aligned} \quad (6)$$

path type 2.

$$\begin{aligned}
S_{p_3}^{\text{dist}}(\mathbf{d}_0, \mathbf{d}_1, \mathbf{d}_2, \mathbf{d}_3) &= \int_{-\infty}^{+\infty} \int_{-\infty}^{h_2} \int_{-\infty}^{h_1} S_3^{\text{dist}}(\mathbf{d}_0, h_0, \mathbf{d}_1, h_1, \mathbf{d}_2, h_2, \mathbf{d}_3) P_{h_1}^1(h_0) dh_0 P_{h_2}^1(h_1) dh_1 P_{+\infty}^1(h_2) dh_2 \\
&= \underbrace{\int_{-\infty}^{+\infty} G_1^{\text{dist}}(\mathbf{d}_3, h_2) \int_{-\infty}^{h_2} G_1^{\text{dist}}(\mathbf{d}_2, h_1, h_2) \underbrace{\int_{-\infty}^{h_1} G_1^{\text{dist}}(\mathbf{d}_1, h_0, h_1) G_1^{\text{dist}}(-\mathbf{d}_0, h_0) P_{h_1}^1(h_0) dh_0 P_{h_2}^1(h_1) dh_1 P_{+\infty}^1(h_2) dh_2}_{M_2}}_{M_1} \\
&\quad \underbrace{\hspace{10em}}_{M_0} \\
&= \int_{-\infty}^{+\infty} (C^1(h_2))^{\Lambda(\mathbf{d}_3)} \int_{-\infty}^{h_2} \left(\frac{C^1(h_1)}{C^1(h_2)} \right)^{\Lambda(\mathbf{d}_2)} \int_{-\infty}^{h_1} \left(\frac{C^1(h_0)}{C^1(h_1)} \right)^{\Lambda(\mathbf{d}_1)} (C^1(h_0))^{\Lambda(-\mathbf{d}_0)} P_{h_1}^1(h_0) dh_0 P_{h_2}^1(h_1) dh_1 P_{+\infty}^1(h_2) dh_2 \\
&= \left(\frac{1}{1 + \Lambda(\mathbf{d}_1) + \Lambda(-\mathbf{d}_0)} \right) \int_{-\infty}^{+\infty} (C^1(h_2))^{\Lambda(\mathbf{d}_3)} \int_{-\infty}^{h_2} \left(\frac{C^1(h_1)}{C^1(h_2)} \right)^{\Lambda(\mathbf{d}_2)} (C^1(h_1))^{\Lambda(-\mathbf{d}_0)} P_{h_2}^1(h_1) dh_1 P_{+\infty}^1(h_2) dh_2 \\
&= \left(\frac{1}{1 + \Lambda(\mathbf{d}_1) + \Lambda(-\mathbf{d}_0)} \right) \left(\frac{1}{1 + \Lambda(\mathbf{d}_2) + \Lambda(-\mathbf{d}_0)} \right) \int_{-\infty}^{+\infty} (C^1(h_2))^{\Lambda(\mathbf{d}_3)} (C^1(h_2))^{\Lambda(-\mathbf{d}_0)} P_{+\infty}^1(h_2) dh_2 \\
&= \frac{1}{(1 + \Lambda(\mathbf{d}_1) + \Lambda(-\mathbf{d}_0))(1 + \Lambda(\mathbf{d}_2) + \Lambda(-\mathbf{d}_0))(1 + \Lambda(\mathbf{d}_3) + \Lambda(-\mathbf{d}_0))}. \tag{7}
\end{aligned}$$

path type 3.

$$\begin{aligned}
S_{p_3}^{\text{dist}}(\mathbf{d}_0, \mathbf{d}_1, \mathbf{d}_2, \mathbf{d}_3) &= \underbrace{G_1^{\text{dist}}(\mathbf{d}_3)}_{w_1} \underbrace{G_1^{\text{dist}}(-\mathbf{d}_0) S_{p_2}^{\text{dist}}(\mathbf{d}_1, \mathbf{d}_2, \mathbf{d}_3)}_{\text{forward}} + \underbrace{G_1^{\text{dist}}(-\mathbf{d}_0) G_1^{\text{dist}}(\mathbf{d}_3)}_{w_2} \underbrace{S_{p_2}^{\text{dist}}(-\mathbf{d}_2, -\mathbf{d}_1, -\mathbf{d}_0)}_{\text{inverse}} \\
&= \underbrace{\frac{1}{1 + \Lambda(\mathbf{d}_3)}}_{w_1} \underbrace{\frac{1}{1 + \Lambda(-\mathbf{d}_0)} S_{p_2}^{\text{dist}}(\mathbf{d}_1, \mathbf{d}_2, \mathbf{d}_3)}_{\text{forward}} + \underbrace{\frac{1}{1 + \Lambda(-\mathbf{d}_0)} \frac{1}{1 + \Lambda(\mathbf{d}_3)} S_{p_2}^{\text{dist}}(-\mathbf{d}_2, -\mathbf{d}_1, -\mathbf{d}_0)}_{\text{inverse}} \\
&= \frac{1}{(1 + \Lambda(\mathbf{d}_3))(1 + \Lambda(-\mathbf{d}_0))} \frac{1}{(1 + \Lambda(\mathbf{d}_2) + \Lambda(-\mathbf{d}_1))(1 + \Lambda(\mathbf{d}_3) + \Lambda(-\mathbf{d}_1))} \\
&\quad + \frac{1}{(1 + \Lambda(-\mathbf{d}_0))(1 + \Lambda(\mathbf{d}_3))} \frac{1}{(1 + \Lambda(-\mathbf{d}_1) + \Lambda(\mathbf{d}_2))(1 + \Lambda(-\mathbf{d}_0) + \Lambda(\mathbf{d}_2))} \\
&= \frac{1}{(1 + \Lambda(-\mathbf{d}_0))(1 + \Lambda(\mathbf{d}_3))} \left(\frac{1}{(1 + \Lambda(\mathbf{d}_2) + \Lambda(-\mathbf{d}_1))(1 + \Lambda(\mathbf{d}_3) + \Lambda(-\mathbf{d}_1))} + \frac{1}{(1 + \Lambda(-\mathbf{d}_1) + \Lambda(\mathbf{d}_2))(1 + \Lambda(-\mathbf{d}_0) + \Lambda(\mathbf{d}_2))} \right). \tag{8}
\end{aligned}$$

1.3 Reciprocity of our model

We use the same vertex term as in previous work [Wang et al. 2022], which satisfies reciprocity. Here, we prove the reciprocity of our segment term (Eq. 5).

Given a *path type 1*, $\tilde{x}_{\text{fwd}}^{t_1} = (\mathbf{d}_0, \mathbf{d}_1, \dots, \mathbf{d}_{k-1}, \mathbf{d}_k)$, its position-free segment term is given by Eq. 5:

$$S_{p_k}^{\text{dist}_{t_1 \text{fwd}}}(\mathbf{d}_0, \mathbf{d}_1, \dots, \mathbf{d}_{k-1}, \mathbf{d}_k) = \frac{1}{1 + \Lambda(-\mathbf{d}_0) + \Lambda(\mathbf{d}_k)} \cdot \frac{1}{1 + \Lambda(-\mathbf{d}_1) + \Lambda(\mathbf{d}_k)} \cdots \frac{1}{1 + \Lambda(-\mathbf{d}_{k-2}) + \Lambda(\mathbf{d}_k)} \cdot \frac{1}{1 + \Lambda(-\mathbf{d}_{k-1}) + \Lambda(\mathbf{d}_k)}. \tag{9}$$

Given a *path type 2*, $\bar{x}_{\text{fwd}}^{t2} = (\mathbf{d}_0, \mathbf{d}_1, \dots, \mathbf{d}_{k-1}, \mathbf{d}_k)$, its position-free segment term is given by Eq. 5:

$$S_{p_k}^{\text{dist_t2fwd}}(\mathbf{d}_0, \mathbf{d}_1, \dots, \mathbf{d}_{k-1}, \mathbf{d}_k) = \frac{1}{1 + \Lambda(\mathbf{d}_1) + \Lambda(-\mathbf{d}_0)} \cdot \frac{1}{1 + \Lambda(\mathbf{d}_2) + \Lambda(-\mathbf{d}_0)} \cdots \frac{1}{1 + \Lambda(\mathbf{d}_{k-1}) + \Lambda(-\mathbf{d}_0)} \cdot \frac{1}{1 + \Lambda(\mathbf{d}_k) + \Lambda(-\mathbf{d}_0)} \quad (10)$$

Given a *path type 3*, $\bar{x}_{\text{fwd}}^{t3} = (\mathbf{d}_0, \mathbf{d}_1, \dots, \mathbf{d}_{k-1}, \mathbf{d}_k)$, its position-free segment term is given by Eq. 5:

$$S_{p_k}^{\text{dist_t3fwd}}(\mathbf{d}_0, \mathbf{d}_1, \dots, \mathbf{d}_{k-1}, \mathbf{d}_k) = \frac{1}{(1 + \Lambda(-\mathbf{d}_0))(1 + \Lambda(\mathbf{d}_k))} \left(S_{p_{k-1}}^{\text{dist}}(\mathbf{d}_1, \dots, \mathbf{d}_k) + S_{p_{k-1}}^{\text{dist}}(-\mathbf{d}_{k-1}, \dots, -\mathbf{d}_0) \right). \quad (11)$$

The inverse *path type 1* $\bar{x}_{\text{inv}}^{t1} = (-\mathbf{d}_k, -\mathbf{d}_{k-1}, \dots, -\mathbf{d}_1, -\mathbf{d}_0)$ is *path type 2*. Therefore, its position-free segment term is given by Eq. 10:

$$\begin{aligned} S_{p_k}^{\text{dist_t1inv}}(-\mathbf{d}_k, -\mathbf{d}_{k-1}, \dots, -\mathbf{d}_0) &= \frac{1}{1 + \Lambda(-\mathbf{d}_{k-1}) + \Lambda(\mathbf{d}_k)} \cdot \frac{1}{1 + \Lambda(-\mathbf{d}_{k-2}) + \Lambda(\mathbf{d}_k)} \cdots \frac{1}{1 + \Lambda(-\mathbf{d}_1) + \Lambda(\mathbf{d}_k)} \cdot \frac{1}{1 + \Lambda(-\mathbf{d}_0) + \Lambda(\mathbf{d}_k)} \\ &= S_{p_k}^{\text{dist_t2fwd}}(\mathbf{d}_0, \mathbf{d}_1, \dots, \mathbf{d}_{k-1}, \mathbf{d}_k) \end{aligned} \quad (12)$$

The inverse *path type 2* $\bar{x}_{\text{inv}}^{t2} = (-\mathbf{d}_k, -\mathbf{d}_{k-1}, \dots, -\mathbf{d}_1, -\mathbf{d}_0)$ is *path type 1*. Therefore, its position-free segment term is given by Eq. 9:

$$\begin{aligned} S_{p_k}^{\text{dist_t2inv}}(-\mathbf{d}_k, -\mathbf{d}_{k-1}, \dots, -\mathbf{d}_1, -\mathbf{d}_0) &= \frac{1}{1 + \Lambda(\mathbf{d}_k) + \Lambda(-\mathbf{d}_0)} \cdot \frac{1}{1 + \Lambda(\mathbf{d}_{k-1}) + \Lambda(-\mathbf{d}_0)} \cdots \frac{1}{1 + \Lambda(\mathbf{d}_2) + \Lambda(-\mathbf{d}_0)} \cdot \frac{1}{1 + \Lambda(\mathbf{d}_1) + \Lambda(-\mathbf{d}_0)} \\ &= S_{p_k}^{\text{dist_t3fwd}}(\mathbf{d}_0, \mathbf{d}_1, \dots, \mathbf{d}_{k-1}, \mathbf{d}_k) \end{aligned} \quad (13)$$

The inverse *path type 3* $\bar{x}_{\text{inv}}^{t3} = (-\mathbf{d}_k, -\mathbf{d}_{k-1}, \dots, -\mathbf{d}_1, -\mathbf{d}_0)$ is *path type 3*. Therefore, its position-free segment term is given by Eq. 11:

$$\begin{aligned} S_{p_k}^{\text{dist_t3inv}}(-\mathbf{d}_k, -\mathbf{d}_{k-1}, \dots, -\mathbf{d}_1, -\mathbf{d}_0) &= \frac{1}{(1 + \Lambda(\mathbf{d}_k))(1 + \Lambda(-\mathbf{d}_0))} \left(S_{p_{k-1}}^{\text{dist}}(-\mathbf{d}_{k-1}, \dots, -\mathbf{d}_0) + S_{p_{k-1}}^{\text{dist}}(\mathbf{d}_1, \dots, \mathbf{d}_k) \right) \\ &= S_{p_k}^{\text{dist_t3fwd}}(\mathbf{d}_0, \mathbf{d}_1, \dots, \mathbf{d}_{k-1}, \mathbf{d}_k) \end{aligned} \quad (14)$$

2 Path type-aware BRDF evaluation

In this section, we provide additional details, pseudocode, and further validation for PTABE (path type-aware BRDF evaluation).

2.1 Estimator for BRDF evaluation

PTABE builds upon the unidirectional path tracing estimator proposed by Wang et al. [2022]. The estimator is defined as:

$$\rho(\omega_i, \omega_o) \approx \frac{1}{N} \sum_{j=0}^N \sum_{i=2}^{k+1} e(\bar{x}_i), \quad (15)$$

where $\bar{x}_i = (\mathbf{d}_0, \mathbf{d}_1, \dots, \mathbf{d}_i)$ denotes a path of length i , and $e(\bar{x}_i) = \frac{f(\bar{x}_i)}{\text{pdf}(\bar{x}_i)}$. Here, we denote $e(\bar{x}_i)$ as the sample weight of the path. $f(\bar{x}_i)$ is evaluated according to Eq. 17 in the main paper. The path probability density function $\text{pdf}(\bar{x}_i)$ is computed as the product of the PDFs used to sample the internal directions, from $\mathbf{d}_1$ to $\mathbf{d}_{i-2}$:

$$\text{pdf}(\bar{x}_i) = \prod_{j=0}^{i-3} \text{pdf}(-\mathbf{d}_j, \mathbf{d}_{j+1}), \quad (16)$$

where $\text{pdf}(-\mathbf{d}_j, \mathbf{d}_{j+1})$ denotes the PDF of sampling direction $\mathbf{d}_{j+1}$ given the incoming direction $-\mathbf{d}_j$. The specific sampling method and corresponding PDF values are computed using the distribution of visible normals (VNDF) sampling technique.

2.2 Pseudocode

We provide the pseudocode (Alg. 1) of PTABE here. Compared to the general BRDF evaluation process [Cui et al. 2023; Wang et al. 2022], we additionally introduce a resampling step (highlighted in blue) to convert *path type 1* into *path type 3*, aiming to allocate more samples to *path type 3*, which exhibits a larger increase in variance under RR.

2.3 Validation

2.3.1 Reason for the larger variance increase when applying RR to path type 3. Fig. 7 in the main paper shows that applying RR to *path type 3* introduces a larger increase in variance than applying RR to *path type 1* or *path type 2*. Here, we provide further validation to explain why applying RR to *path type 3* leads to a larger increase in the estimator variance.

In Fig. 1, we illustrate the variance increase of sample weights (Eq. 15) for the three path types per bounce before and after applying RR under different incident angles. Note that we start from the 4th bounce because RR is typically applied from this point onward in BRDF evaluation; earlier bounces have large contributions and therefore do not require RR.

We further present, in Figs. 3–5, the variance of sample weights for the three path types before and after applying RR, which serve to validate the results shown in Fig. 1.

As observed in Fig. 1, applying RR to *path type 1* often results in a larger increase in the variance of its sample weights than applying RR to *path type 3*. However, Fig. 2 indicates that the probability of *path type 1* is significantly lower than that of *path type 3*. Therefore, applying RR to *path type 3* leads to a larger increase in the overall estimator variance than applying RR to *path type 1* (Fig. 7 in the main paper).

2.3.2 Performance of PTABE with and without RR. In Fig. 6, we present the variance of our models (without PTABE, with and without RR; and with PTABE, with and without RR) at a maximum bounce of 10. We observe that the variance of our model with PTABE (without RR) is nearly identical to that of our model without PTABE (without RR). Without applying RR, allocating more samples to *path type 3* has little effect on reducing the estimator variance.

However, when RR is applied, our model with PTABE (with RR) achieves significantly lower variance than our model without PTABE (with RR). This demonstrates that PTABE effectively mitigates the variance increase introduced by RR.

2.3.3 Effectiveness of the two heuristics. The resampling step in PTABE alters the path distribution in a non-probabilistic manner, introducing bias. To mitigate this bias, we adopt two simple heuristics. Heuristic 1 performs resampling with probability $1 - P_d$. Heuristic 2 weights the resampled *path type 3* by $1/(1 - P_d)$.

To validate the effectiveness of these two heuristics, we first consider four variants of PTABE: PTABE without heuristics 1 and 2 (Alg. 2); PTABE with heuristic 1 (Alg. 3); PTABE with heuristic 2 (Alg. 4); and PTABE with both heuristics 1 and 2 (Alg. 1). Note that PTABE with both heuristics 1 and 2 (Alg. 1) corresponds to the version used in the main paper.

In Fig. 7, we compare our models on a sphere scene with isotropic copper (GGX, $\alpha = 1.0$) under direct lighting, including: without PTABE, with PTABE (w/o heuristics 1 and 2), with PTABE (w/ heuristic 1), with PTABE (w/ heuristic 2), and with PTABE (w/ heuristics 1 and 2). All results are rendered with 1024 spp and are fully converged. We compute the MSE with respect to the ground truth rendered by Heitz et al. [2016]. The MSE of PTABE without heuristics 1 and 2

indicates that naively converting all *path type 1* into *path type 3* introduces bias. The results for PTABE with heuristic 1 and with heuristic 2 show that each heuristic individually mitigates the bias introduced by path conversion. Finally, the results for PTABE with both heuristics demonstrate that their effects are complementary, leading to further MSE reduction.

In Fig. 8, we show the mean reflectance of our models—without PTABE, with PTABE (w/o heuristics 1 and 2), with PTABE (w/ heuristic 1), with PTABE (w/ heuristic 2), and with PTABE (w/ heuristics 1 and 2)—under different maximum bounce counts and incident angles. To better highlight the differences among PTABE variants, we also plot the reflectance differences between each PTABE variant and the baseline model without PTABE in Fig. 9. We observe that converting *path type 1* into *path type 3* at a maximum bounce of 3 generally leads to overestimation, which can be explained by Figs. 10 and 11. Using the heuristics further increases this overestimation. However, for maximum bounce counts greater than 3, path conversion leads to underestimation. The PTABE variants with heuristics compensate for this underestimation, resulting in better agreement with the baseline as the maximum bounce increases. PTABE with both heuristics shows the best overall performance among the tested variants. These results demonstrate the effectiveness of the two heuristics.

In Fig. 10, we further present the mean reflectance per bounce for our models—without PTABE and with PTABE variants (w/o heuristics 1 and 2, w/ heuristic 1, w/ heuristic 2, and w/ heuristics 1 and 2)—under different incident angles. We observe that, at the 3rd bounce, PTABE generally leads to overestimation. At higher bounce counts, however, it tends to underestimate the reflectance, among which the variant with both heuristics performs best. From this perspective, the overestimation at the 3rd bounce and the underestimation at subsequent bounces balance each other, allowing PTABE with both heuristics to achieve reasonable results when the number of bounces is sufficiently large. The per-bounce reflectance behavior further explains the variation of reflectance with respect to the maximum bounce shown in Fig. 8.

In Fig. 11, we show the mean sample weight (Eq. 15) per bounce for the three path types of our model without PTABE under different incident angles. The relative magnitudes of the sample weights for path types 1 and 3 help explain the overestimation and underestimation observed after converting *path type 1* into *path type 3* in Fig. 10. There are two notable cases. First, at the 3rd bounce with an incident angle of 80° , the sample weight of *path type 1* is significantly larger than that of *path type 3*. However, after path conversion, the reflectance at the 3rd bounce shows only a slight underestimation (Fig. 10). This is because the probability of *path type 1* is very low at grazing angles (Fig. 2), meaning that path conversion occurs infrequently. Second, at the 6th bounce, the sample weight of *path type 3* is significantly larger than that of *path type 1*. Nevertheless, the reflectance at the 6th bounce still exhibits underestimation after path conversion. This is because the reflectance at the 6th bounce also depends on that of the 5th bounce, which is already substantially underestimated. In addition, the contribution of paths at the 6th bounce is relatively small, so the overestimation introduced by path conversion is insufficient to compensate for the accumulated underestimation from previous bounces.

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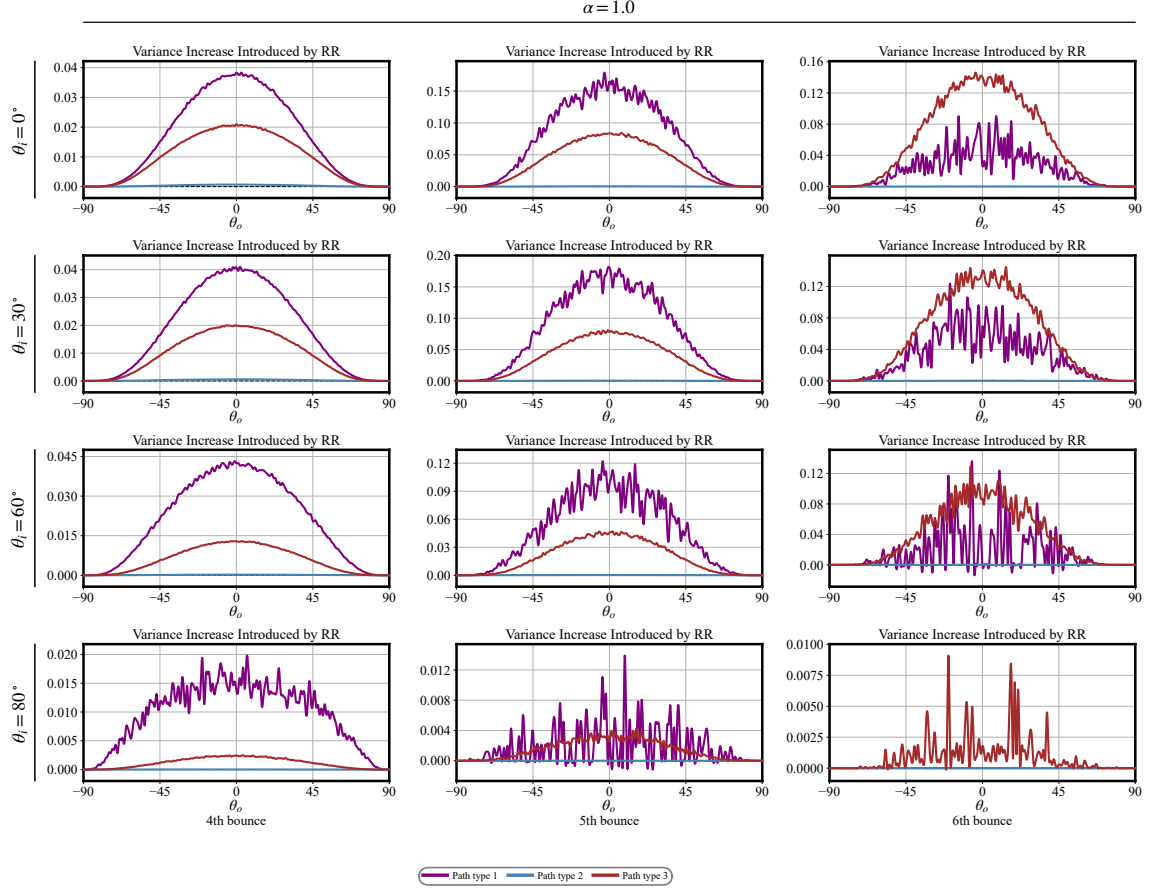


Fig. 1. Variance increase of sample weights (Eq. 15) for the three path types per bounce before and after applying RR under different incident angles. Note that all results are generated using our model without PTABE. Although *path type 1* often exhibits a larger per-path variance increase, its low probability (Fig. 2) leads to a smaller contribution to the overall estimator variance compared to *path type 3*.

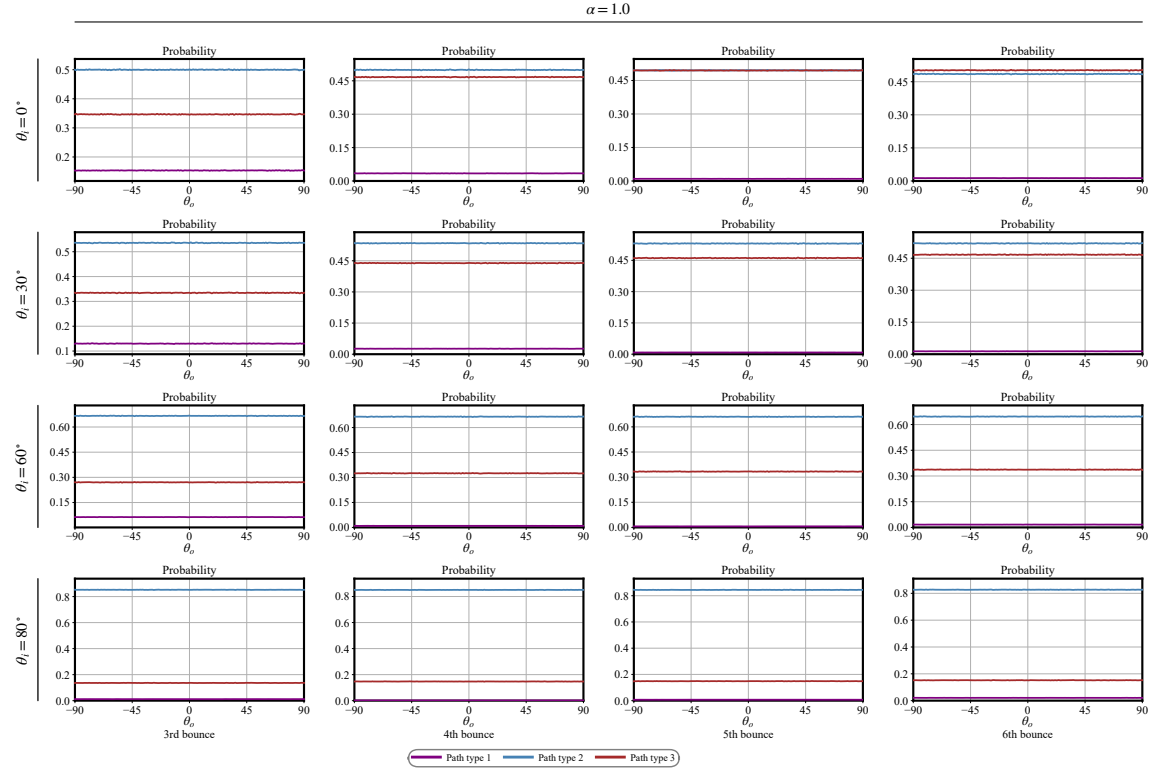


Fig. 2. Probability of the three path types of our model without PTABE given the path probability density function (Eq. 16) under different incident angles. The results show that *path type 1* has a significantly lower probability than *path type 3*.

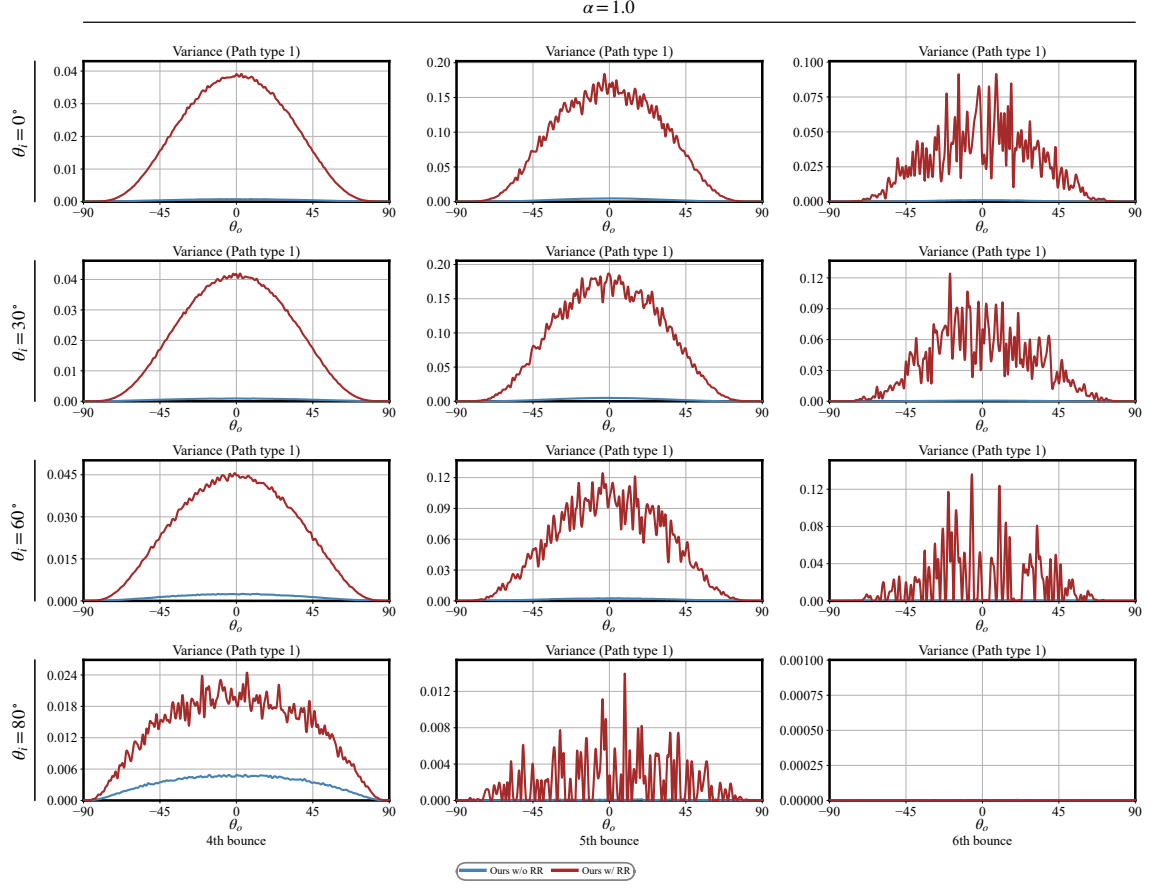


Fig. 3. Variance of sample weights for *path type 1* of our model without PTABE before and after applying RR. These results validate the variance increase trends observed in Fig. 1.

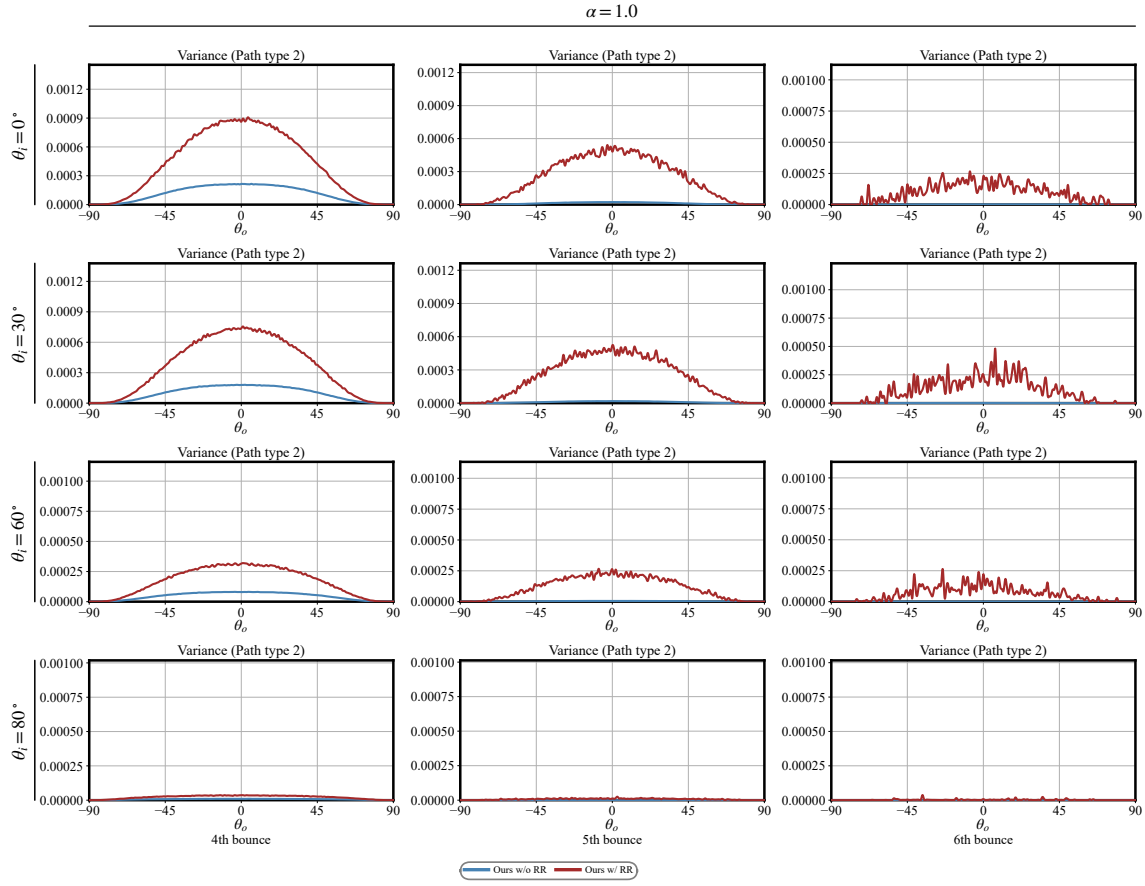


Fig. 4. Variance of sample weights for *path type 2* of our model without PTABE before and after applying RR. These results validate the variance increase trends observed in Fig. 1.

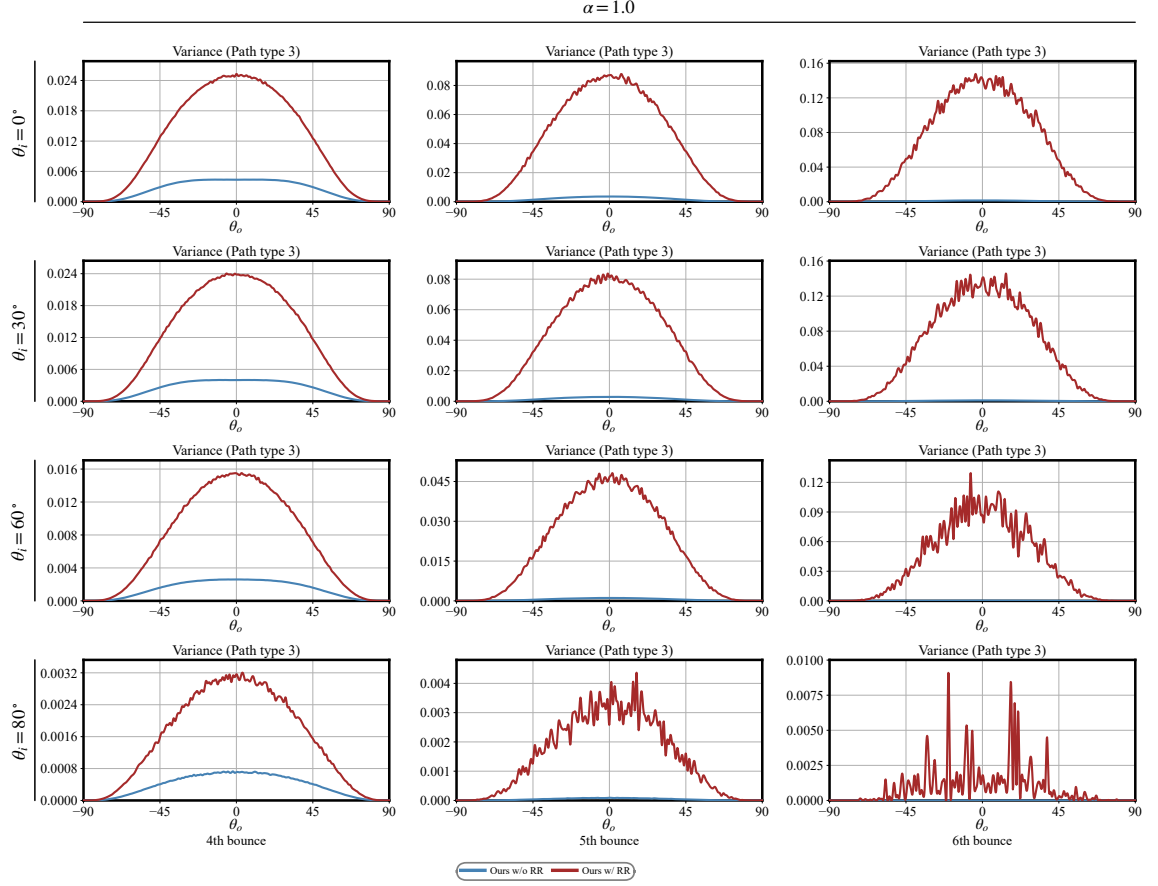


Fig. 5. Variance of sample weights for path type 3 of our model without PTABE before and after applying RR. These results validate the variance increase trends observed in Fig. 1.

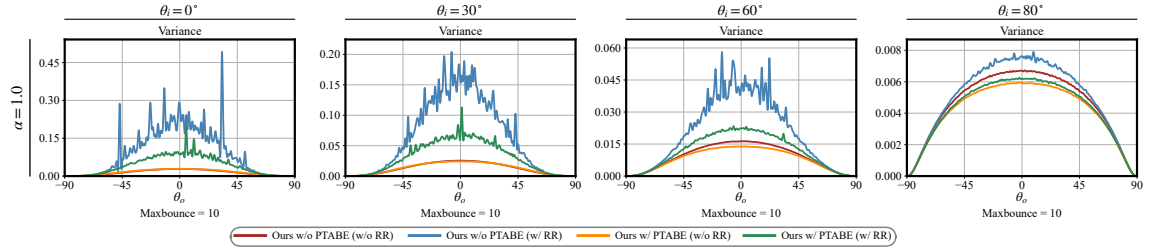


Fig. 6. Variance comparison of our models with and without PTABE, under both RR and non-RR settings at a maximum bounce of 10. Without RR, PTABE shows negligible impact on variance. With RR, PTABE significantly reduces variance, suggesting its effectiveness in mitigating the variance increase introduced by RR.

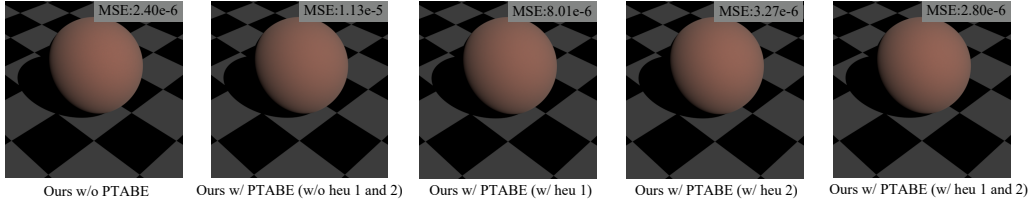


Fig. 7. Comparison of our models on a sphere scene with isotropic copper (GGX, $\alpha = 1.0$) under direct lighting, including: without PTABE, with PTABE (w/o heuristics 1 and 2), with PTABE (w/ heuristic 1), with PTABE (w/ heuristic 2), and with PTABE (w/ heuristics 1 and 2). All results are rendered with 1024 spp and are fully converged. PTABE without heuristics 1 and 2 shows clear bias due to naive path-type conversion. Applying either heuristic 1 or heuristic 2 individually reduces this bias, while combining both heuristics yields the lowest MSE, demonstrating their complementary effects.

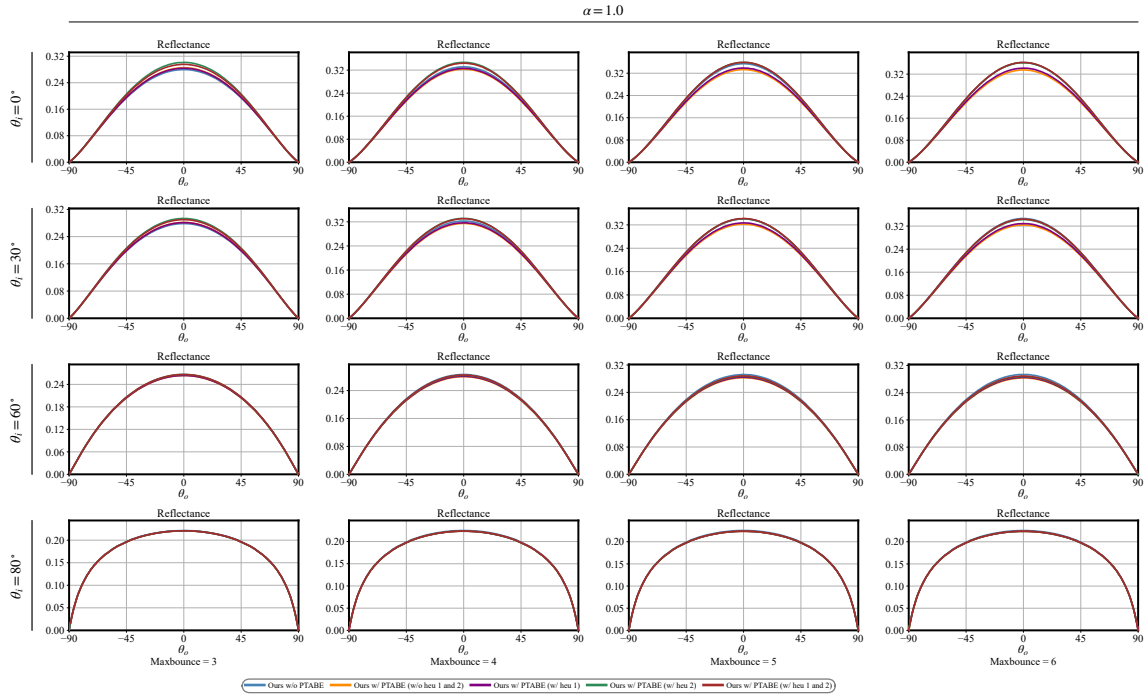


Fig. 8. Mean reflectance of our models—without PTABE and with PTABE variants (w/o heuristics 1 and 2, w/ heuristic 1, w/ heuristic 2, and w/ heuristics 1 and 2)—under varying maximum bounce counts and incident angles. Fig. 9 shows the reflectance differences between each PTABE variant and the model without PTABE.

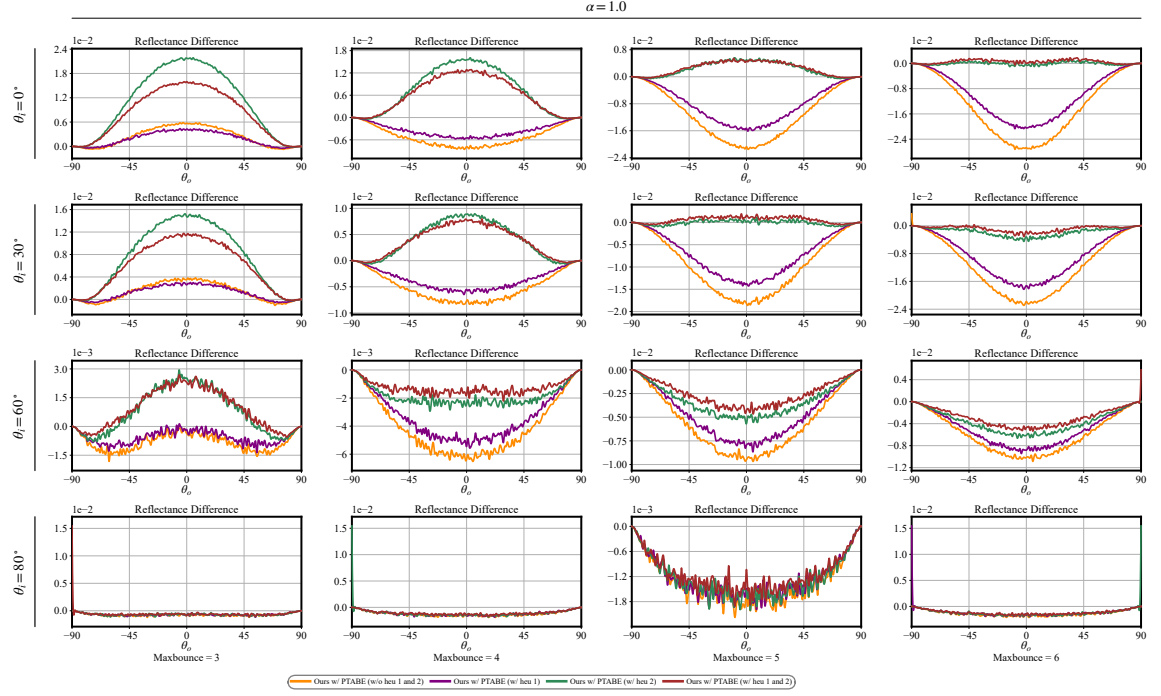


Fig. 9. Reflectance differences between PTABE variants and the model without PTABE. The results highlight converting *path type 1* into *path type 3* at a maximum bounce of 3 generally leads to overestimation. However, when the maximum bounce exceeds 3, path conversion instead results in underestimation. Two heuristics help compensate this underestimation, leading to closer agreement with the baseline as the maximum bounce increases.

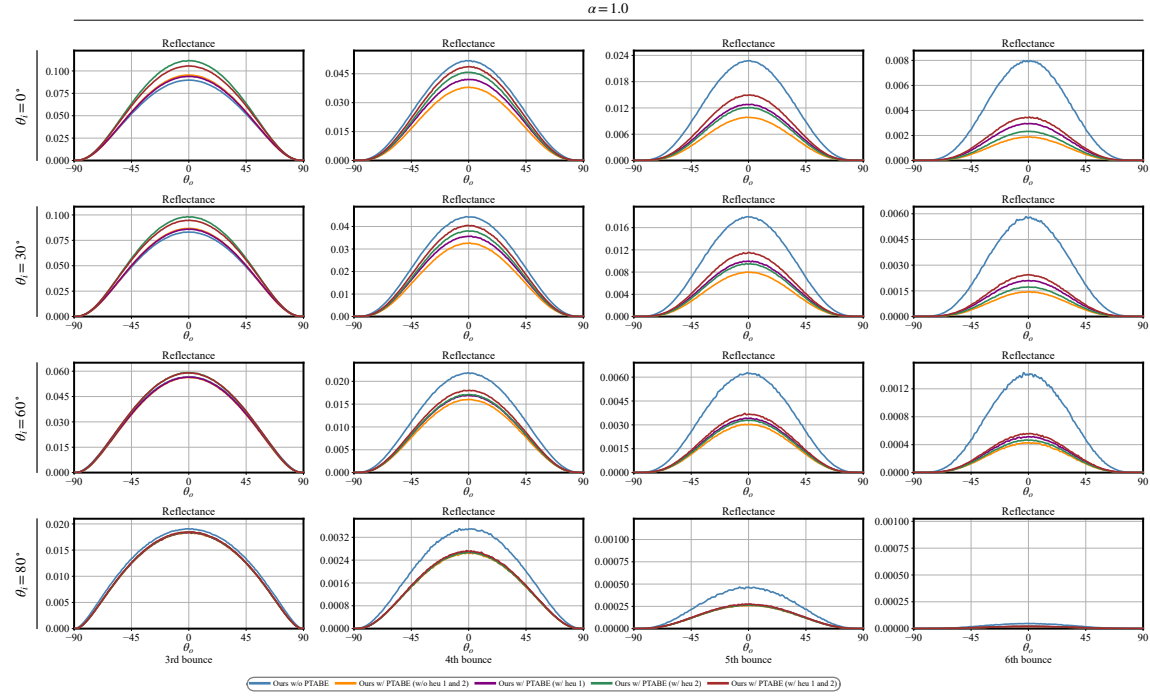


Fig. 10. Mean reflectance per bounce for our models—without PTABE and with PTABE variants (w/o heuristics 1 and 2, w/ heuristic 1, w/ heuristic 2, and w/ heuristics 1 and 2)—under different incident angles. PTABE tends to overestimate at the 3rd bounce but underestimate at higher bounces. The variant with both heuristics performs best, as these opposing biases balance out over multiple bounces, leading to better agreement with the baseline and explaining the trends observed in Fig. 8.

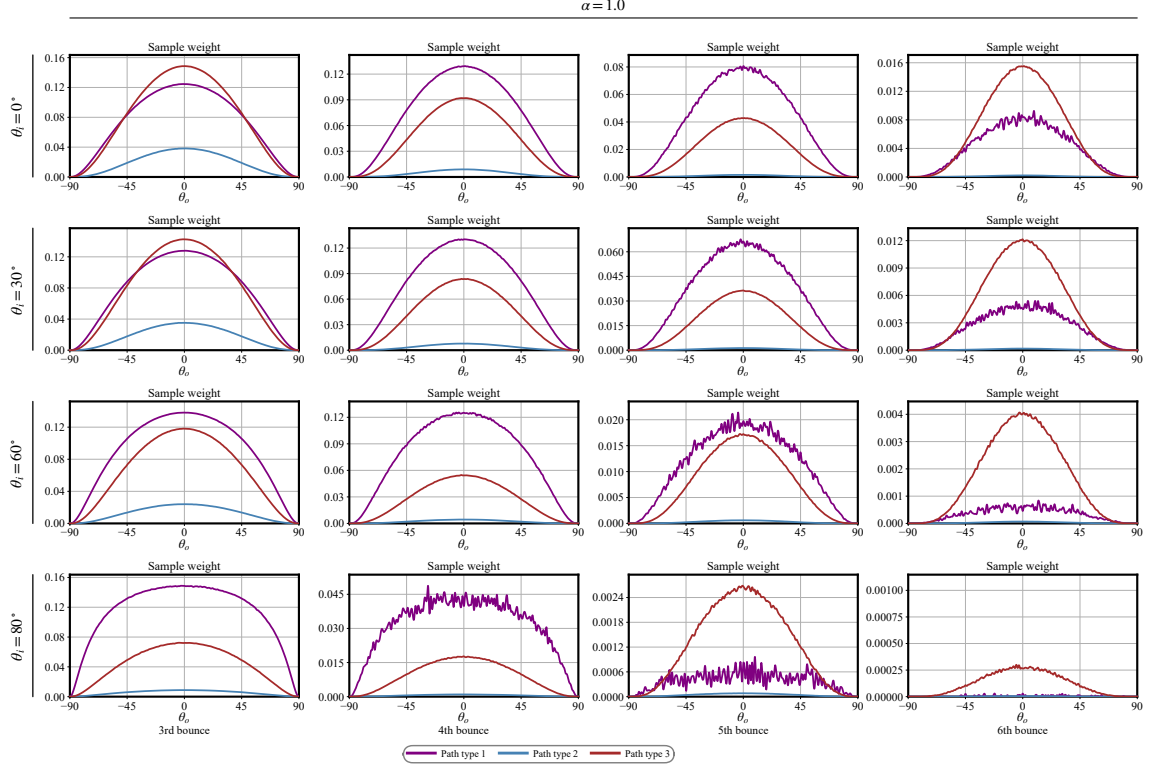


Fig. 11. Mean sample weight (Eq. 15) per bounce for the three path types of our model without PTABE under different incident angles. The relative magnitudes of path types 1 and 3 explain the overestimation at low bounces and underestimation at higher bounces observed after path conversion (Fig. 10).

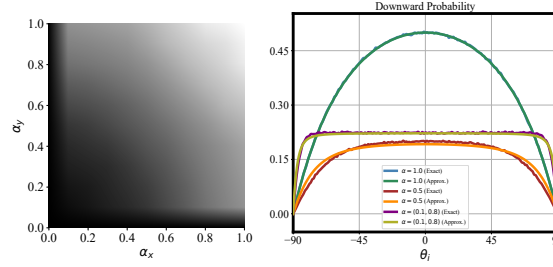


Fig. 12. $T(\alpha_x, \alpha_y)$ is precomputed by comparing $G_1(\omega_i)$ with the exact probability of downward reflection (left). $P_d(\omega_i)$ is a good approximation of the probability of downward reflection (right).

ALGORITHM 1: Path type-aware BRDF evaluation (w/ heu 1 and 2)

```

1 function eval( $\omega_i, \omega_o$ )
2   SegmentTerm  $s(\Lambda(\omega_o))$  // initializing;
3    $f \leftarrow 0$  // result;
4    $k \leftarrow 0$  // bounce;
5    $w \leftarrow 1$  // weight;
6    $\mathbf{d}_0 \leftarrow -\omega_i$ ;
7    $s.addBounce(\Lambda(\mathbf{d}_0))$ ;
8    $f \leftarrow f + w \cdot v(\mathbf{d}_0, \omega_o) \cdot s.getS_k()$  // single bounce;
9   for  $k \leftarrow 1 \dots maxBounce - 1$  do
10     $\mathbf{d}_k, p_{k-1} \leftarrow sample(\mathbf{d}_{k-1})$ ;
11    if  $k > 1$  and  $\mathbf{d}_{k-1}.z < 0$  and  $\mathbf{d}_k.z < 0$  // path type 1 then
12       $q_d \leftarrow P_d(\mathbf{d}_{k-1})$  // downward probability;
13      if  $rand() > q_d$  // heuristic 1 then
14         $\mathbf{d}_k, p_{k-1} \leftarrow sample(\mathbf{d}_{k-1})$ ;
15        if  $\mathbf{d}_k.z > 0$  // path type 3 then
16           $w \leftarrow \frac{w}{1 - q_d}$  // weight path type 3 (heuristic 2);
17      if  $\mathbf{d}_k.z < 0$  // path type 1 then
18        if  $rand() > q_d$  // heuristic 1 then
19           $\mathbf{d}_k.z \leftarrow -\mathbf{d}_k.z$  // force it to path type 3 ;
20           $w \leftarrow \frac{w}{1 - q_d}$  // weight path type 3 (heuristic 2);
21     $s.addBounce(\Lambda(\mathbf{d}_k))$ ;
22     $w \leftarrow \frac{w \cdot v(\mathbf{d}_{k-1}, \mathbf{d}_k)}{p_{k-1}}$  // vertex term and pdf;
23     $S_k \leftarrow s.getS_k()$ ;
24    if  $k \geq rrDepth$  then
25       $q \leftarrow S_k$ ;
26      if  $rand() > q$  then
27        break
28       $w \leftarrow \frac{w}{q}$  // russian roulette;
29     $f \leftarrow f + w \cdot v(\mathbf{d}_k, \omega_o) \cdot s.getS_k()$ ;
30  return  $f$ 

```

ALGORITHM 2: Path type-aware BRDF evaluation (w/o heu 1 and 2)

```

1 function eval( $\omega_i, \omega_o$ )
2   SegmentTerm  $s(\Lambda(\omega_o))$  // initializing;
3    $f \leftarrow 0$  // result;
4    $k \leftarrow 0$  // bounce;
5    $w \leftarrow 1$  // weight;
6    $\mathbf{d}_0 \leftarrow -\omega_i$ ;
7    $s.addBounce(\Lambda(\mathbf{d}_0))$ ;
8    $f \leftarrow f + w \cdot v(\mathbf{d}_0, \omega_o) \cdot s.getS_k()$  // single bounce;
9   for  $k \leftarrow 1 \dots maxBounce - 1$  do
10     $\mathbf{d}_k, p_{k-1} \leftarrow sample(\mathbf{d}_{k-1})$ ;
11    if  $k > 1$  and  $\mathbf{d}_{k-1}.z < 0$  and  $\mathbf{d}_k.z < 0$  // path type 1 then
12       $q_d \leftarrow P_d(\mathbf{d}_{k-1})$  // downward probability;
13       $\mathbf{d}_k, p_{k-1} \leftarrow sample(\mathbf{d}_{k-1})$ ;
14      if  $\mathbf{d}_k.z < 0$  // path type 1 then
15         $\mathbf{d}_k.z \leftarrow -\mathbf{d}_k.z$  // force it to path type 3;
16     $s.addBounce(\Lambda(\mathbf{d}_k))$ ;
17     $w \leftarrow \frac{w \cdot v(\mathbf{d}_{k-1}, \mathbf{d}_k)}{p_{k-1}}$  // vertex term and pdf;
18     $S_k \leftarrow s.getS_k()$ ;
19    if  $k \geq rrDepth$  then
20       $q \leftarrow S_k$ ;
21      if  $rand() > q$  then
22        break
23       $w \leftarrow \frac{w}{q}$  // russian roulette;
24     $f \leftarrow f + w \cdot v(\mathbf{d}_k, \omega_o) \cdot s.getS_k()$ ;
25  return  $f$ 

```

ALGORITHM 3: Path type-aware BRDF evaluation (w/ heu 1)

```

1 function eval( $\omega_i, \omega_o$ )
2   SegmentTerm  $s(\Lambda(\omega_o))$  // initializing;
3    $f \leftarrow 0$  // result;
4    $k \leftarrow 0$  // bounce;
5    $w \leftarrow 1$  // weight;
6    $\mathbf{d}_0 \leftarrow -\omega_i$ ;
7    $s.addBounce(\Lambda(\mathbf{d}_0))$ ;
8    $f \leftarrow f + w \cdot v(\mathbf{d}_0, \omega_o) \cdot s.getS_k()$  // single bounce;
9   for  $k \leftarrow 1 \dots maxBounce - 1$  do
10     $\mathbf{d}_k, p_{k-1} \leftarrow sample(\mathbf{d}_{k-1})$ ;
11    if  $k > 1$  and  $\mathbf{d}_{k-1}.z < 0$  and  $\mathbf{d}_k.z < 0$  // path type 1 then
12       $q_d \leftarrow P_d(\mathbf{d}_{k-1})$  // downward probability;
13      if  $rand() > q_d$  // heuristic 1 then
14         $\mathbf{d}_k, p_{k-1} \leftarrow sample(\mathbf{d}_{k-1})$ ;
15      if  $\mathbf{d}_k.z < 0$  // path type 1 then
16        if  $rand() > q_d$  // heuristic 1 then
17           $\mathbf{d}_k.z \leftarrow -\mathbf{d}_k.z$  // force it to path type 3;
18     $s.addBounce(\Lambda(\mathbf{d}_k))$ ;
19     $w \leftarrow \frac{w \cdot v(\mathbf{d}_{k-1}, \mathbf{d}_k)}{p_{k-1}}$  // vertex term and pdf;
20     $S_k \leftarrow s.getS_k()$ ;
21    if  $k \geq rrDepth$  then
22       $q \leftarrow S_k$ ;
23      if  $rand() > q$  then
24        break
25       $w \leftarrow \frac{w}{q}$  // russian roulette;
26     $f \leftarrow f + w \cdot v(\mathbf{d}_k, \omega_o) \cdot s.getS_k()$ ;
27 return  $f$ 

```

ALGORITHM 4: Path type-aware BRDF evaluation (w/ heu 2)

```

1 function eval( $\omega_i, \omega_o$ )
2   SegmentTerm  $s(\Lambda(\omega_o))$  // initializing;
3    $f \leftarrow 0$  // result;
4    $k \leftarrow 0$  // bounce;
5    $w \leftarrow 1$  // weight;
6    $\mathbf{d}_0 \leftarrow -\omega_i$ ;
7    $s.addBounce(\Lambda(\mathbf{d}_0))$ ;
8    $f \leftarrow f + w \cdot v(\mathbf{d}_0, \omega_o) \cdot s.getS_k()$  // single bounce;
9   for  $k \leftarrow 1 \dots maxBounce - 1$  do
10     $\mathbf{d}_k, p_{k-1} \leftarrow sample(\mathbf{d}_{k-1})$ ;
11    if  $k > 1$  and  $\mathbf{d}_{k-1}.z < 0$  and  $\mathbf{d}_k.z < 0$  // path type 1 then
12       $q_d \leftarrow P_d(\mathbf{d}_{k-1})$  // downward probability;
13       $\mathbf{d}_k, p_{k-1} \leftarrow sample(\mathbf{d}_{k-1})$ ;
14      if  $\mathbf{d}_k.z > 0$  // path type 3 then
15         $w \leftarrow \frac{w}{1 - q_d}$  // weight path type 3 (heuristic 2);
16      if  $\mathbf{d}_k.z < 0$  // path type 1 then
17         $\mathbf{d}_k.z \leftarrow -\mathbf{d}_k.z$  // force it to path type 3;
18         $w \leftarrow \frac{w}{1 - q_d}$  // weight path type 3 (heuristic 2);
19     $s.addBounce(\Lambda(\mathbf{d}_k))$ ;
20     $w \leftarrow \frac{w \cdot v(\mathbf{d}_{k-1}, \mathbf{d}_k)}{p_{k-1}}$  // vertex term and pdf;
21     $S_k \leftarrow s.getS_k()$ ;
22    if  $k \geq rrDepth$  then
23       $q \leftarrow S_k$ ;
24      if  $rand() > q$  then
25        break
26       $w \leftarrow \frac{w}{q}$  // russian roulette;
27     $f \leftarrow f + w \cdot v(\mathbf{d}_k, \omega_o) \cdot s.getS_k()$ ;
28  return  $f$ 

```
